Spyder

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C:\Users\wamul\OneDrive\Desktop\Computer-Logic-and-Programming\untitled7.py

untitled6.py | untitled7.py*

```python
# -*- coding: utf-8 -*-
"""
Created on Mon Apr  6 13:07:51 2026

@author: wamul
# Module 14 Problems - Classes & Objects
# Problem 1: Employee Class with Bonus Method
"""

class Employee:
    def __init__(self, name, id_number, department, salary):
        self.name = name
        self.id_number = id_number
        self.department = department
        self.salary = salary

    def display_info(self):
        print(f"\nEmployee Name:   {self.name}")
        print(f"ID Number:       {self.id_number}")
        print(f"Department:      {self.department}")
        print(f"Salary:          ${self.salary:,.2f}")

    def compute_bonus(self, bonus_rate):
        """Compute and return employee bonus based on rate x salary."""
        return bonus_rate * self.salary


print("=" * 50)
print("  PROBLEM 1 - Employee Class with Bonus")
print("=" * 50)

emp1 = Employee("Susan Meyers", "47899", "Accounting", 50000.00)
emp1.display_info()

rate = 0.10   # 10% bonus rate
bonus = emp1.compute_bonus(rate)
print(f"Bonus Rate:      {rate * 100:.0f}%")
print(f"Bonus Amount:    ${bonus:,.2f}")

emp2 = Employee("Mark Jones", "39119", "IT", 72000.00)
emp2.display_info()
```

Console 1/A

```
==================================================
  PROBLEM 1 - Employee Class with Bonus
==================================================

Employee Name:   Susan Meyers
ID Number:       47899
Department:      Accounting
Salary:          $50,000.00
Bonus Rate:      10%
Bonus Amount:    $5,000.00

Employee Name:   Mark Jones
ID Number:       39119
Department:      IT
Salary:          $72,000.00
Bonus Rate:      15%
Bonus Amount:    $10,800.00

--------------------------------------------------
```

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```python
rate2 = 0.15  # 15% bonus rate
bonus2 = emp2.compute_bonus(rate2)
print(f"Bonus Rate:      {rate2 * 100:.0f}%")
print(f"Bonus Amount:    ${bonus2:,.2f}")


# ==========================================================
# Problem 2: Student Class with Tuition Method
# ==========================================================

class Student:
    def __init__(self, first_name, last_name, district_code, enrolled_credits):
        self.first_name = first_name
        self.last_name = last_name
        self.district_code = district_code.upper()
        self.enrolled_credits = enrolled_credits

    def compute_tuition(self):
        """Compute tuition: $250/credit in-district, $500/credit out-of-district."""
        if self.district_code == "I":
            rate = 250.00
        else:
            rate = 500.00
        return self.enrolled_credits * rate

    def display_info(self):
        tuition = self.compute_tuition()
        print(f"\nStudent:         {self.first_name} {self.last_name}")
        print(f"District Code:   {self.district_code}")
        print(f"Enrolled Credits:{self.enrolled_credits}")
        print(f"Tuition Owed:    ${tuition:,.2f}")


print("\n" + "=" * 50)
print("  PROBLEM 2 - Student Class with Tuition")
print("=" * 50)

s1 = Student("Alice", "Johnson", "I", 12)
s1.display_info()

s2 = Student("Bob", "Smith", "O", 9)
s2.display_info()
```

Console 1/A

```
Salary:          $72,000.00
Bonus Rate:      15%
Bonus Amount:    $10,800.00

==================================================
  PROBLEM 2 - Student Class with Tuition
==================================================

Student:         Alice Johnson
District Code:   I
Enrolled Credits:12
Tuition Owed:    $3,000.00

Student:         Bob Smith
District Code:   O
Enrolled Credits:9
Tuition Owed:    $4,500.00
```

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```python
# ==========================================================
# Problem 3: Student Class with Dictionary for District Codes
# ==========================================================

class StudentV2:
    # Class-level dictionary: district code -> tuition rate per credit hour
    TUITION_RATES = {
        "I": 250.00,   # In-district
        "O": 500.00,   # Out-of-district
        "X": 800.00,   # International
        "G": 250.00,   # Reciprocity (same as in-district)
    }

    def __init__(self, first_name, last_name, district_code, enrolled_credits):
        self.first_name = first_name
        self.last_name = last_name
        self.district_code = district_code.upper()
        self.enrolled_credits = enrolled_credits

    def compute_tuition(self):
        """Compute tuition using the district code dictionary.
        Defaults to out-of-district rate if code is unrecognized."""
        rate = self.TUITION_RATES.get(self.district_code, 500.00)
        return self.enrolled_credits * rate

    def display_info(self):
        tuition = self.compute_tuition()
        rate = self.TUITION_RATES.get(self.district_code, 500.00)
        descriptions = {
            "I": "In-District",
            "O": "Out-of-District",
            "X": "International",
            "G": "Reciprocity",
        }
        desc = descriptions.get(self.district_code, "Unknown")
        print(f"\nStudent:        {self.first_name} {self.last_name}")
        print(f"District Code:   {self.district_code} ({desc})")
        print(f"Rate/Credit:     ${rate:,.2f}")
        print(f"Enrolled Credits:{self.enrolled_credits}")
        print(f"Tuition Owed:    ${tuition:,.2f}")
```

untitled7.py
- Employee
  - __init__
  - display_info
  - compute_bo
- Student
- StudentV2

Console 1/A

```
==================================================
  PROBLEM 3 - Student Class with Dictionary
==================================================

Student:        Alice Johnson
District Code:   I (In-District)
Rate/Credit:     $250.00
Enrolled Credits:12
Tuition Owed:    $3,000.00

Student:        Bob Smith
District Code:   O (Out-of-District)
Rate/Credit:     $500.00
Enrolled Credits:9
Tuition Owed:    $4,500.00

Student:        Chen Wei
District Code:   X (International)
```

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```python
    def compute_tuition(self):
        """Compute tuition using the district code dictionary.
        Defaults to out-of-district rate if code is unrecognized."""
        rate = self.TUITION_RATES.get(self.district_code, 500.00)
        return self.enrolled_credits * rate

    def display_info(self):
        tuition = self.compute_tuition()
        rate = self.TUITION_RATES.get(self.district_code, 500.00)
        descriptions = {
            "I": "In-District",
            "O": "Out-of-District",
            "X": "International",
            "G": "Reciprocity",
        }
        desc = descriptions.get(self.district_code, "Unknown")
        print(f"\nStudent:        {self.first_name} {self.last_name}")
        print(f"District Code:   {self.district_code} ({desc})")
        print(f"Rate/Credit:     ${rate:,.2f}")
        print(f"Enrolled Credits:{self.enrolled_credits}")
        print(f"Tuition Owed:    ${tuition:,.2f}")


print("\n" + "=" * 50)
print("  PROBLEM 3 - Student Class with Dictionary")
print("=" * 50)

students = [
    StudentV2("Alice",   "Johnson",  "I", 12),   # In-district
    StudentV2("Bob",     "Smith",    "O",  9),   # Out-of-district
    StudentV2("Chen",    "Wei",      "X", 15),   # International
    StudentV2("Maria",   "Garcia",   "G",  6),   # Reciprocity
]

for student in students:
    student.display_info()

print("\n" + "=" * 50)
print("  All programs completed successfully.")
print("=" * 50)
```

Console 1/A

```
Enrolled Credits:9
Tuition Owed:    $4,500.00

Student:        Chen Wei
District Code:   X (International)
Rate/Credit:     $800.00
Enrolled Credits:15
Tuition Owed:    $12,000.00

Student:        Maria Garcia
District Code:   G (Reciprocity)
Rate/Credit:     $250.00
Enrolled Credits:6
Tuition Owed:    $1,500.00

==================================================
  All programs completed successfully.
==================================================
```